Vehicle Sales between 2002-2023

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Forecasting Methods – Professors Bruno Damásio and Carolina Vasconcelos

Introduction

In this study, we examine vehicle sales from 2002 to 2023. Our analysis aims to uncover the intricate patterns and influences driving automotive consumption over this long timeframe. By thoroughly examining these extensive sales trends, we hope to provide a more nuanced understanding of the changing dynamics in the automotive industry.

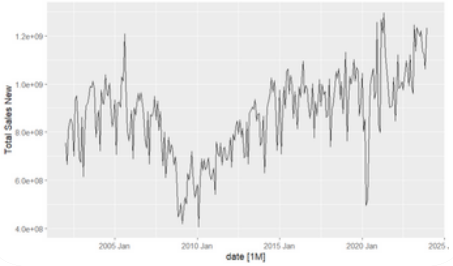
Data

Our analysis was based on a Kaggle dataset, containing 6 variables and 264 observations. With this dataset we were able to find valuable insights into the trends and patterns of vehicle sales over this century, allowing us to understand better the dynamics of the automotive market and make informed decisions about the forecasting of the next years in the vehicle sales business.

Firstly we had to merge the first two columns of the original dataset in order to get the index of the time series, pass from the columns "Year" and "Month" to "Date". Our study was based on the total number of new vehicles sold through the years.

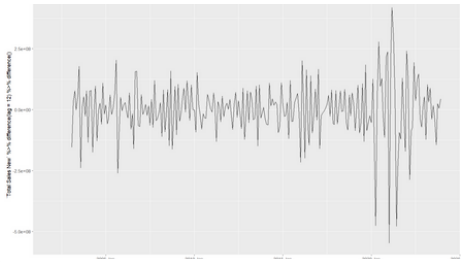
Transformations

Now, we needed to check if the variance was constant, if there were any trends, or if we had cyclic data:



There is a decreasing trend from 2002 to 2010, and a increasing trend from there until 2023. Besides that, the variance is not constant and we also have seasonality.
With the previous information, we needed to apply some transformations to get a stationary series.
We applied the square root, logarithm and BoxCox transformations, but the differentiation was the most effective one. Firstly, to remove the seasonal effects of the data we differentiated with a lag of 12 periods, then, we used a differentiation of consecutive values. To check if there was a need to make more differentiations, we could prove that we could reject the null hypothesis.

With this transformations we were able to get a more stationary series.



Model Selection

After checking the ACF and PACF we were able to identify a seasonal MA(1) model - even though there are various lags significantly different from zero we can see that the lag=12, is where there are a higher differenciation, and that its neighbours are the result of it. By analysing the rest of the lags we could say that lag=2, lag=7 or lag=9 could also be took into account in the model selection.

With this information, we fitted several models, using the Information Criteria procedure. The results said that the best models were ARIMA(2,1,2)(0,1,2)[12], ARIMA(1,1,2)(0,1,2)[12] and ARIMA(0,1,2)(0,1,2)[12], with the first one being the best.

$$\Delta Y_t = -0.3882 \cdot Y_{t-1} + 0.2202 \cdot Y_{t-2} + \epsilon_t + 0.1049 \cdot \epsilon_{t-1} - 0.6845 \cdot \epsilon_{t-2} - 0.7830 \cdot \Delta\epsilon_{t-1} - 0.0691 \cdot \Delta\epsilon_{t-2}$$

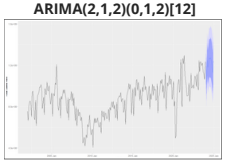
Checking the residuals: While looking to the residuals' ACF, it was noted that certain lags retained significantly different from zero. This suggests that our model may not fully capture all the underlying patterns present in the data, despite exhaustive efforts to optimize the model parameters.

Ljung-box statistic test: With this test we were able to conclude that, since the p-value of all of the models is less than the commonly used significance level of 0.05, we reject the null hypothesis. This suggests that there is significant autocorrelation in the residuals of the models at a lag of 12.

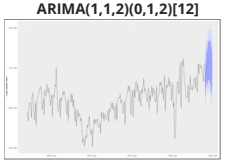
After several different testings on the transformations of the data, and after looking to all of the possibilities of a SARIMA model, we can say that given the dataset's constraints and the analytical tools at our disposal, the chosen model represents the best compromise between model complexity and predictive performance. While the identified model may not be perfect, it demonstrates a careful and thorough approach to time series analysis. Recognising the possibility of p-hacking increases the transparency of our methodology, ensuring that the reported results are robust and reliable in the context of the analysis.

Forecasting

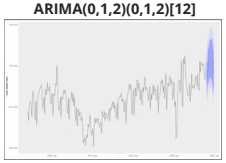
ARIMA(2,1,2)(0,1,2)[12]



ARIMA(1,1,2)(0,1,2)[12]



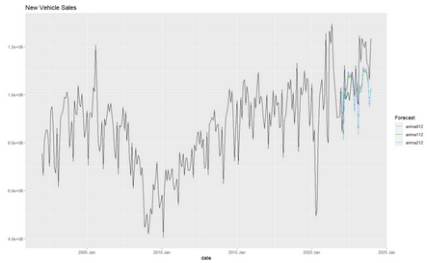
ARIMA(0,1,2)(0,1,2)[12]



We can see that the forecasts for each model are very similar to each other.

Accuracy

Here, we divided our data between train (2002-2021) and test (2022-2023).



.model	.type	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE	ACF1
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
arima012	Test	55511360.	113859136.	99835228.	4.48	9.20	1.12	0.844	0.333
arima112	Test	62314989.	115652919.	99518967.	5.13	9.13	1.11	0.857	0.324
arima212	Test	51090406.	111706423.	98056592.	4.06	9.05	1.10	0.828	0.342

The **ARIMA212** model had the best score in every performance measure. Therefore, it seems to be clearly the most suitable choice.

Conclusion and Reflection

In conclusion, this project's exhaustive exploration of forecasting methods yielded valuable insights into the predictive modeling of new vehicle sales. Through rigorous experimentation with various data transformations and SARIMA model configurations, we have developed a set of models that strike a balance between complexity and prediction accuracy. While acknowledging the inherent limitations of any forecasting effort, we have attempted to maintain transparency and rigor throughout our analysis, avoiding the traps of p-hacking and ensuring the robustness of our reported results.
By leveraging the forecasts generated by our top-performing models, we anticipate a gradual increase in new vehicle sales over the next year, although with the typical fluctuations of the automotive market. Despite the inherent uncertainty, our chosen model provides useful guidance to decision-makers navigating this dynamic landscape. We have worked hard to provide a solid foundation for strategic planning and decision-making in the automotive industry by paying close attention to methodological detail and emphasizing empirical validation.